



Figure 1: Typical technology functions of an organisation in supporting business

it. They keep minimal functioning software for free and market it heavily. They end up confusing developers and organisations, who think they have adopted OSS based application development and give into the ease-

of-use hype, but end up paying large sums of licensing fees down the road. Rewriting the application to get out of using the software takes time and money, which most organisations don't have when they are growing. This is

exactly why there should be a focus on design and architecture while selecting the right open source platform, software, and tools in developing applications from the ground up.

Business functions of an organisation

The first step in developing technology strategy is to understand the business functions of the organisation. Once the functions are broken down, the listing and consolidation of the core technical capabilities of competitors will provide a holistic view of what's needed to get a tech organisation up and running.

A typical organisation consists of many business functions; however, many of these do not need software development. One needs to evaluate if a business function will act as a

Business function	Core capability	Open source software(s)
E-commerce and mobile app	Online shopping portal on the Web and on mobile devices working across multiple popular browsers and mobile operating systems	Open Cart, WooCommerce, Zen Cart, Shuup, Ubercart, Spree Commerce. PhoneGap, Flutter, Ionic, BuildFire
Customer relationship management	Managing customer information and interactions across all touch points	SuiteCRM, openCRX, EspoCRM, vTiger, Zurmo
Customer service	Customer support, request tracking and management	Bugzilla, Helpy, osTicket
Marketing	Promoting and selling products and services using emails and campaigns, and generating leads with tracking	Geary, Mautic, OpenEMM, Thunderbird
Document and content management	Website and portals for internal use, customers, investors, etc; Documents management and file sharing	WordPress, Drupal, Joomla, OpenKM, 7Zip, WinSCP, FileZilla, FireFTP
Productivity tools	Creating text documents, presentations, spreadsheets	Apache OpenOffice, LibreOffice
Monitoring and dashboard reporting	Tools to collect, analyse and report data, and develop visualisation charts with dashboards and notifications	Graphana, Datawrapper, Charted, Draw.io, D3
Accounting and finance	Managing resources and assets, order and invoice management, compliance reporting	Apache OFBiz, ERPNext, Tryton, GNU Cash, OpenBravo, GanttProject
Operating system (OS) and networking	Platform to run applications and connect across applications, tools and end-point devices	Linux, Ubuntu, Nagios

Table 1